

FIRST FIT

INPUT:

```
kaviya@localhost:~/firstfit — /usr/bin/vim ff.c
#include<stdio.h>
int main(){
    int m,n;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
    int blocksize[m], remainingsize[m];
    printf("Enter size of each block:\n");
    for(int i=0;i<m;i++){
        scanf("%d", &blocksize[i]);
        remainingsize[i]=blocksize[i];
    }
    printf("Enter number of processess:\n");
    scanf("%d", &n);
    int processsize[n];
    printf("Enter size of each process:\n");
    for(int i=0;i<n;i++){
        scanf("%d", &processsize[i]);
    }
    for(int i=0;i<n;i++){
        int allocated=0;
        for(int j=0;j<m;j++){
            if(remainingsize[j] >= processsize[i]){
                int fragment = remainingsize[j]-processsize[i];
                1,8
                Top
            }
        }
    }
}
```

```
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    for(int i=0;i<n;i++){
        scanf("%d", &processsize[i]);
    }
    for(int i=0;i<n;i++){
        int allocated=0;
        for(int j=0;j<m;j++){
            if(remainingsize[j] >= processsize[i]){
                int fragment = remainingsize[j]-processsize[i];
                printf("Process %d of size %d is allocated to block %d of size %d with fragment %d\n", i+1, processsize[i], j+1, blocksize[j], fragment);
                remainingsize[j] -= processsize[i];
                allocated = 1;
                break;
            }
        }
        if(!allocated){
            printf("Process %d of size %d is not allocated\n", i+1, processsize[i]);
        }
    }
    return 0;
}
35,1
Bot
```

OUTPUT:

```
kaviya@localhost:~/firstfit
[kaviya@localhost ~]$ cd firstfit
[kaviya@localhost firstfit]$ vi ff.c
[kaviya@localhost firstfit]$ gcc ff.c
[kaviya@localhost firstfit]$ ./a.out
Enter number of memory blocks:
2
Enter size of each block:
150 300
Enter number of processess:
3
Enter size of each process:
50 70 130
Process 1 of size 50 is allocated to block 1 of size 150 with fragment 100
Process 2 of size 70 is allocated to block 1 of size 150 with fragment 30
Process 3 of size 130 is allocated to block 2 of size 300 with fragment 170
[kaviya@localhost firstfit]$
```

BEST FIT

INPUT:

```
kaviya@localhost:~/bestfit — /usr/bin/vim bf.c
#include<stdio.h>
int main(){
    int m,n;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
    int blocksize[m], remaining[m];
    printf("Enter size of each block:\n");
    for(int i=0;i<m;i++){
        scanf("%d", &blocksize[i]);
        remaining[i] = blocksize[i];
    }
    printf("Enter number of processes:\n");
    scanf("%d", &n);
    int processsize[n];
    printf("Enter size of each process:\n");
    for(int i=0;i<n;i++){
        scanf("%d", &processsize[i]);
    }
    for(int i=0;i<n;i++){
        int bestindex = -1;
        for(int j=0;j<m;j++){
            if(remaining[j] >= processsize[i]){
                @@@
                1,1
                Top
```

```
                if(remaining[j] >= processsize[i]){
                    if(bestindex == -1 || remaining[j] < remaining[bestindex]){
                        bestindex = j;
                    }
                    if(bestindex != -1){
                        int frag = remaining[bestindex] - processsize[i];
                        printf("Process %d of size %d is allocated to block %d of size %d with fragmentation %d\n", i+1, processsize[i], bestindex+1, blocksize[bestindex], frag);
                        remaining[bestindex] -= processsize[i];
                    }
                    else
                    {
                        printf("Process %d of size %d is not allocated\n", i+1, processsize[i]);
                    }
                }
            }
        }
    }
    return 0;
}
```

34, 1-8 95%

OUTPUT:

```
kaviya@localhost:~/bestfit
[kaviya@localhost ~]$ cd bestfit
[kaviya@localhost bestfit]$ vi bf.c
[kaviya@localhost bestfit]$ gcc bf.c
[kaviya@localhost bestfit]$ ./a.out
Enter number of memory blocks:
2
Enter size of each block:
150 300
Enter number of processes:
3
Enter size of each process:
30 70 150
Process 1 of size 30 is allocate to block 1 of size 150 with fragement 120
Process 1 of size 30 is allocate to block 1 of size 150 with fragement 90
Process 2 of size 70 is allocate to block 1 of size 150 with fragement 20
Process 2 of size 70 is allocate to block 1 of size 150 with fragement -50
Process 3 of size 150 is allocate to block 2 of size 300 with fragement 150
[kaviya@localhost bestfit]$
```